

[ASRJC/2024/Q3]

3 Name your Jupyter Notebook as:

TASK3_<your name>_<centre number>_<index number>.ipynb

QuickBuy is an online retailer that wants to improve its customer shopping experience. Customers can leave reviews and ratings for products they have purchased.

QuickBuy needs to keep track of its data using both a relational database (SQLite) and a NoSQL database (MongoDB) and object-oriented programming techniques.

The following shows the relational schema for the SQLite database `quickbuy.db` with three tables:

- `customers` (customer_id, customer_name, email)
 - `products` (product_id, product_name, category)
 - `orders` (customer_id#, product_id#, order_date, quantity)
- # denotes foreign key

Task 3.1

Write program code to create the three tables for the relational database. [6]

Task 3.2

Write program code to insert records from `customers.txt`, `products.txt` and `orders.txt` into the three tables. [5]

Task 3.3

Write program code to get and output the total quantity of products ordered by each customer. [3]

Task 3.4

Write program code to get and output the number of orders and total quantity for each product category for orders made in the year 2023. [4]

Task 3.5

Write program code to insert data from `reviews.json` to a `quickbuy` MongoDB database `reviews` collection. [3]

Task 3.6

Write program code to find and output all highly rated products (at least rating 4) with qualitative reviews. [3]

Task 3.7

Write program code to find and output all the reviews for product P5 and calculate and output its average rating correct to 1 decimal place. [4]

Save your Jupyter Notebook for Task 3.